

Computer Architecture
Chapter - 03 (part - 2)

(NTB)

Floating Point addition: (same for subtraction)

$$\# 35.23142 + 0.00053$$

step 1: Decimal to Binary and Normalize

$$a = 35.23142$$

$$= .100011 \cdot 0011101111 \text{ (Binary)}$$

$$= 1.000110011101111 \times 2^5 \text{ (Normalized)}$$

$$b = 0.00053$$

$$= 0.000000000010001011 \text{ (Binary)}$$

$$= 1.0001011 \times 2^{-11} \text{ (Normalized)}$$

Step-2: Match the lower exponent with the higher exponent.

$$a = 1.000110011101111 \times 2^5$$

$$b = 1.0001011 \times 2^{-11}$$

$$= 0.0000000000000000000010001011 \times 2^5$$

Step-3: Add

$$a+b = \left(\begin{array}{l} 1.000110011101111 + \\ 0.0000000000000000000010001011 \end{array} \right) \times 2^5$$

$$= 1.0001100111011110001011 \times 2^5$$

$$= 100011.00111011110001011$$

$$= 35.2342224121 \text{ (Decimal)}$$

Floating point Multiplication:

$$5.234 \times (-0.003)$$

Step-01: Decimal to Binary and Normalize

$$a = 5.234$$

$$= 101.0011101111$$

$$1.010011101111 \times 2^2$$

$$b = 0.003 = 0.000000011000100101$$

$$1.1000100101 \times 2^{-9}$$

Step-02: Do Multiplication

$$a \times b = (1.010011101111 \times 2^2) \times (1.000100101 \times 2^{-9})$$

$$= (1.010011101111 \times 1.000100101) \times 2^{-7}$$

$$= 10.000000101 \times 2^{-7}$$

$$= 0.0000010000000101 \times 2^{-7}$$

$$= 0.0157012939$$

$$\text{Ans: } -0.0157012939$$

Overflow/underflow Concept:

Overflow/Underflow is detected by the range of the exponent field of floating point representation.

Suppose n bits are allocated for exponent field.

$\therefore$ range for bias exponent: 0 to $(2^n - 1)$

as 0 and $(2^n - 1)$ are reserved

$\therefore$ actual range: 1 to $(2^n - 1) - 1$

$\hookrightarrow$ 1 to $(2^n - 2)$
 $\downarrow$ $\downarrow$
a b
(lower range) (upper range)

Suppose, $P \rightarrow$ a number to check

if $(a \leq P \leq b) \rightarrow$ No overflow/underflow

if $(P < a) \rightarrow$ Underflow

if $(P > b) \rightarrow$ overflow